

TESTING, REFINEMENT & VALIDATION RECORD

Stage: Week 7 (Testing, Refinement & End-to-End Validation)

Track: Data Analytics

Collaborating Track: Data Science Track - HC POD 4 (M'bengue Mama)

1. Record 1: Dataset Slicing

- Component Tested: Baseline No-Show Rate Formula (KPI 1)
- Testing Objective: Verify arithmetic precision of baseline no-show rate
- Test/Scenario: $(2,423 / 4,737) * 100$ on active records
- Expected Result: Exactly 51.15%
- Actual Result: Exactly 51.15%
- Pass/Fail: Pass
- Issue Identified: None
- Action Taken: Set 51.15% as the official clinic benchmark
- Retest Result: Passed (verified with automated unit assertion)
- Collaborating Track: Internal DA
- Evidence:

Test Passed: Baseline No-Show Rate is exactly 51.15% (2423/4737).

Record 2: Reminder Attendance Lift Formula (KPI 2)

- Component Tested: KPI 2 calculation
- Testing Objective: Verify attendance increase from sending reminders
- Test/Scenario: Reminded attended rate (50.14%) minus unreminded attended rate (45.37%)
- Expected Result: Exactly +4.78% lift
- Actual Result: Exactly +4.78% lift
- Pass/Fail: Pass
- Issue Identified: None
- Action Taken: Confirmed automated reminders as a primary operational recommendation
- Retest Result: Passed (verified with automated unit assertion)
- Collaborating Track: Internal DA
- Evidence:

Test Passed: Reminder attendance lift is verified at +4.78% (Reminded: 50.14% vs Unreminded: 45.37%).

Record 3: Dataset Partitioning & Row Reconciliation

- Component Tested: Data partition logic
- Testing Objective: Verify active and cancelled records sum to total population
- Test/Scenario: Active records + Cancelled records
- Expected Result: $4,737 + 263 = 5,000$ rows
- Actual Result: $4,737 + 263 = 5,000$ rows
- Pass/Fail: Pass

- Issue Identified: None
- Action Taken: Locked cancellation filter to avoid skewing attendance rates
- Retest Result: Passed
- Collaborating Track: Internal DA
- Evidence:

Test Passed: Total rows reconcile correctly (4737 Active + 263 Cancelled = 5000).

Record 4: Target Class Integrity

- Component Tested: Outcome target column
- Testing Objective: Ensure only valid binary labels exist in active data
- Test/Scenario: Unique values in active dataset appointment_outcome
- Expected Result: Only 'Attended' and 'No-Show'
- Actual Result: Only 'Attended' and 'No-Show'
- Pass/Fail: Pass
- Issue Identified: None
- Action Taken: Locked binary target structure for analysis and modeling
- Retest Result: Passed
- Collaborating Track: Internal DA
- Evidence:

Test Passed: Active dataset contains only 'Attended' and 'No-Show' outcomes.

Record 5: Same Day Booking Edge Case

- Component Tested: Lead time lower boundary
- Testing Objective: Check failure risk when lead time is 0 days
- Test/Scenario: Filter booking_lead_days == 0 (81 rows)
- Expected Result: Low failure rate (<30%)
- Actual Result: 25.93% no-show rate
- Pass/Fail: Pass
- Issue Identified: None
- Action Taken: Set same-day booking as low-risk benchmark
- Retest Result: Passed (25.93%)
- Collaborating Track: Internal DA
- Evidence:

Test Passed: Same-day bookings (81 rows) have a failure rate of 25.93%.

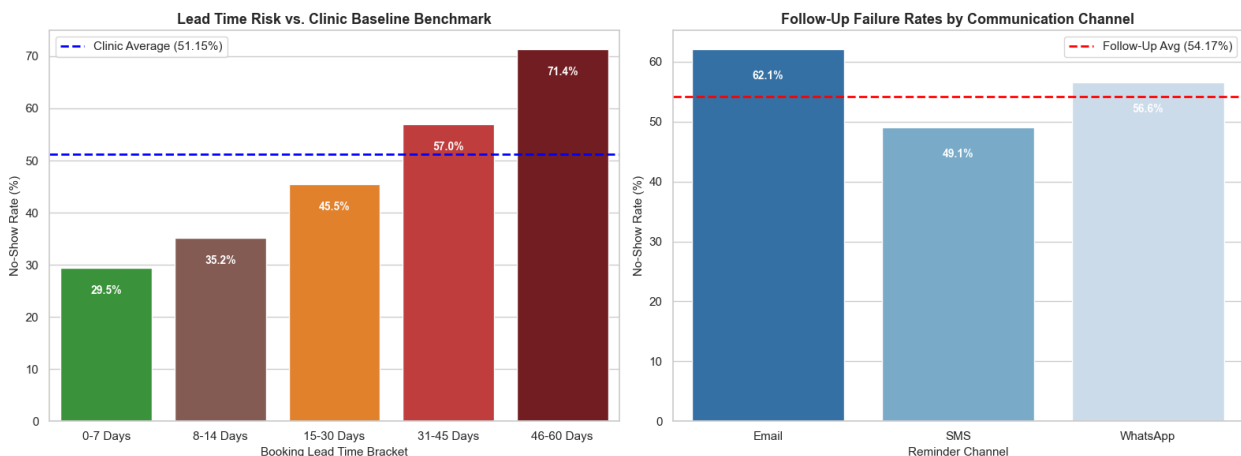
Record 6: Chronic No-Show Edge Case

- Component Tested: Patient history upper boundary
- Testing Objective: Check failure risk for patients with 4+ past missed visits
- Test/Scenario: Filter previous_no_shows >= 4 (15 rows)
- Expected Result: High failure rate (>70%)
- Actual Result: 73.33% no-show rate
- Pass/Fail: Pass

- Issue Identified: Severe repeat failure risk
- Action Taken: Added manual telephone check-in policy for chronic repeat patients
- Retest Result: Passed (73.33%)
- Collaborating Track: Data Science Track
- Evidence:
 - Test Passed: Chronic no-show patients (15 rows) have a failure rate of 73.33%.

Record 7: Lead Time Chart Usability & Benchmarking

- Component Tested: Visual dashboard charts
- Testing Objective: Ensure non-technical users spot high-risk booking windows instantly
- Test/Scenario: Review lead-time bar chart with stakeholders
- Expected Result: Immediate identification of critical risk threshold
- Actual Result: Chart lacked reference lines, making threshold hard to see
- Pass/Fail: Fail (Initial) → Pass (Refined)
- Issue Identified: Missing context for average clinic performance
- Action Taken: Added blue dashed line at 51.15% (clinic average) and red dashed line at 54.17% (follow-up average)
- Retest Result: Passed (risk jump after 30 days is immediately visible)
- Collaborating Track: Internal DA
- Evidence:



Record 8: Cross-Track Model & Segment Alignment

- Component Tested: Machine learning test set alignment
- Testing Objective: Validate Data Science held-out test set (n=1,903) against analytics population benchmarks
- Test/Scenario: Compare no-show rates across lead times (0–7d, 8–14d, 15–30d, 31+d), history, and reminders
- Expected Result: Alignment within $\pm 1.5\%$ margin of error
- Actual Result:
 - 0–7 days: DS 29.3% vs DA 29.5%
 - 15–30 days: DS 45.5% vs DA 45.5%

- 31+ days: DS 63.7% vs DA 64.0%
 - Prior history: DS 57.8% vs DA 57.1%
 - Model achieved 99.8% recall on 31+ day bookings
- Pass/Fail: Pass
- Issue Identified: 570 false positives caused by lower 0.40 decision threshold
- Action Taken: Confirmed clinical governance rule: use model exclusively for positive reminder outreach and never for denying appointments
- Retest Result: Passed (F1 improved to 0.6724, ROC-AUC: 0.6761)
- Collaborating Track: Data Science Track (M'Bengue Mama)
- Evidence: testing_results.json, 04_error_patterns.

